a) **Cellulose** : It is the main constituent of plant cell walls. It is found in all vegetables, fruits and legumes.

b) **Hemicellulose:** It is the main constituent of cereal fibres.It is also present in all vegetables and hulls of legumes.

c) **Pectins:** Pectins are formed by the combination of large number of galacturonic acid molecules. In the presence of sucrose and citric acid, pectin forms a gel. It is used in the food industry as an ingredient of jams and jellies.

d) **Gums and mucilages:** They are non-structural components of plant cells that are soluble in hot water. They are used as additives and stabilizers by the food industry.

e) **Lignin**: They are the tough, woody parts of plants.

Dietary fibers can also be classified as soluble and insoluble fibers based on their solubility in water. Insoluble fibers do not dissolve in water(Cellulose,hemi-cellulose and lignin) and soluble fibres readily dissolve in water(Pectins, gums and mucilages). The types, sources and action of fibre in the body is summarized in **table 9.3**

Food for thought — choose wisely!



What's most important is the type of carbohydrate you choose to eat because some sources are healthier than others. The amount of carbohydrate in the diet – high or low – is less important than the type of carbohydrate in the diet. For example, healthy, whole grains and whole wheat bread, rye and barley are better choices than highly refined white bread or french fries.

Foods high in carbohydrates are an important part of a healthy diet. Carbohydrates provide the body with glucose, which is converted to energy used to support bodily functions and physical activity. But carbohydrate quality is important; some types of carbohydrate-rich foods are better than others:

Table 9.3 Types, Sources and Action of fiber

Types of fiber	Major food sources	Action in the body
Soluble fibers Gums, pectins, mucilages	Citrus fruits, apple, oats, barley, legumes	• Delay gastro intestinal transit • Delay glucose absorption • Lower blood cholesterol
Insoluble fibers Cellulose Hemicellulose	Whole wheat products, wheat bran, whole grain breads, cereals and vegetables like green peas, beans, cabbage. Skin of vegetable and fruits, grains	• Accelerate Gastro intestinal transit • Increase faecal weight • Slow starch hydrolysis • Delay glucose absorption

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